# BIPUL PADHY

(817) 716-9571 | bkp736@eid.utexas.edu | linkedin.com/in/bipulpadhy

## EDUCATION

**The University of Texas at Austin** **May 2028**
**B.S., Electrical and Computer Engineering**
**Overall GPA:** 3.93/4.00
**Honors:** ECE Honors & Engineering Honors Program
**Relevant Coursework:**
**ECE 438:** CMOS Analog Integrated Circuit Design, Amplifiers, Frequency Response, Stability, Miller Compensation
**ECE 462L:** Power Electronics, Loss Analysis, Soft Switching, Small-Signal Modeling and Control, Magnetics Analysis/Design
**ECE 369:** Three Phase Power Systems, Transformers, Power Flow, Fault Analysis, Power System Controls
**ECE 325:** Transmission Lines, Signal Integrity, Wave Propagation, Magnetic Circuits, Skin Depth, Shielding, Maxwell's Equations

## RESEARCH PROJECTS

**Dr. Alex Hanson's Power Electronics and Magnetics Group** **Jan 2026 - Present**
**Undergraduate Research Assistant**

**A Phase-Locked Loop based Multi-MHz Synchronous Rectifier with Inherent Delay Cancellation and ZVS Control**

- Designed a PLL-based synchronous rectifier for 6.78/13.56 MHz wireless power transfer in collaboration with a PhD candidate, modeling PLL/VCO control loops in LTspice and PLECS to align voltage and current phase
- Diagnosed and resolved board-level switching faults using oscilloscope measurements, validating zero-voltage switching in a multi-MHz synchronous rectifier
- Validated AC/DC conversion efficiency of 94-95% at 50 W (6.78 MHz) and approximately 93% at 30 W (13.56 MHz)

**Wireless Coil Parameter Estimation on a SiC Inverter**

- Developed embedded C/C++ firmware and a current-sensing algorithm to estimate wireless-coil self/mutual inductance and secondary-side load resistance in real time
- Designed a 1 kW, 100 kHz SiC inverter PCB layout, optimizing power-device placement, gate-drive routing, and thermal paths for efficient switching operation

## LEADERSHIP & STUDENT GROUPS

**Texas Guadaloop (Hyperloop R&D Team)**
**Power Electronics Lead** **May 2026 - Present**

- Architected a 250 kHz wireless power transfer system across an 8-inch air gap within an $800 budget for dynamic charging of the Gen V Hyperloop pod
- Led design of an uninterruptible power supply for pod levitation, integrating wireless power transfer and lithium-ion battery backup to maintain power availability
- Led Gen IV Hyperloop pod power electronics design, contributing to 1st place in Engineering Design at European Hyperloop Week, an international competition

**Power Electronics Engineer** **Sep 2025 - May 2026**

- Designed, assembled, and validated a multi-rail power-distribution PCB using buck regulators and an LDO to supply 12 V, 5 V, and 3.3 V rails
- Developed and tested a 400 W, 60 V-to-24 V DC/DC power-supply PCB, optimizing component placement and thermal vias for heat dissipation and reliability
- Built and verified a 600 V power system with inverter inrush-current protection, limiting peak startup current to 0.2 A

## TECHNICAL SKILLS

**Hardware Design:** PCB Layout, DC/DC Converters, Buck Regulators, LDOs, Inverters, Gate-Drive Routing, Analog Circuit Analysis
**Validation & Lab:** Board-Level Debugging, Efficiency Characterization, Oscilloscopes, DMMs, Network Analyzers, SMD Soldering
**Simulation & CAD:** Altium Designer, LTspice, PLECS, Cadence, MATLAB
**Programming:** C, C++, Python, Java, Embedded Firmware